View Review

Paper ID

383

Paper Title

AI-Powered Crop Detection & Advisory System using Deep Learning

REVIEW QUESTIONS

1. Relevance of the Work with the theme of Conference ?

Yes

2. Novelty in the Paper

Yes

3. Comments to authors

The paper presents an AI-powered crop disease detection and advisory system using MobileNetV2, achieving over 90% accuracy on the Plant Village dataset. The development
of a user-friendly web and mobile platform for real-time diagnostics and treatment recommendations is a strong practical contribution, particularly for farmers. The methodology, leveraging transfer learning and data augmentation, is well-executed, and the performance metrics (e.g., 0.9948 precision for Maize Common Rust) demonstrate robustness. However, there are areas for improvement. The introduction mistakenly discusses cultural heritage preservation, which seems unrelated to the paper's focus on crop disease detection. This section needs revision to align with the agricultural theme. The literature review is comprehensive but could better highlight how the proposed system advances beyond cited works (e.g., [11], [19]). The dataset description lacks details on the number of images per class, which is critical for assessing class imbalance handling. Additionally, the paper repeats the "Results" section (N duplicates M), which should be consolidated. The discussion on future work (e.g., drone integration) is promising but vague;
specific challenges or planned approaches would strengthen this section. I recommend revising the introduction to focus on agriculture, clarifying dataset details (e.g., class distribution), consolidating redundant sections, and elaborating on how the system improves upon prior work. Including a brief discussion on computational efficiency (e.g., inference time on edge devices) would enhance the paper's relevance for real-world deployment. Overall, the work is promising and practical but requires minor revisions for clarity and focus.

4. Overall Score- Strongly Accept, Accept, Reject or Borderline

Accept

5. Reviewers Confidence

High